

Victor Gardin

Aerospace Engineering Graduate Student – U.S. Citizen
Seeking GNC Engineering opportunities in orbital mechanics & mission analysis, Spring/Summer 2027.

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Education

Master of Science in Aerospace Engineering 2025–2027

Georgia Institute of Technology – Metz, France & Atlanta, USA

Relevant coursework: Optimal Guidance & Control, Orbital Mechanics (ongoing), Machine Learning for Robotics

Arts et Métiers Institute of Technology 2023–2026

Lille, France

Combined BSc & MSc in Mechanical and Industrial Engineering, dual-degree with Georgia Tech.

Coursework: Solid Mechanics, Fluid Dynamics, Power Electronics, CAD. Admitted via a 2-year Advanced Physics & Technology Preparatory Class (Versailles) to this top-10 French engineering school.

Research & Work Experience

Mission Analysis & GNC Engineering Intern Jan–Jun 2026

Agema Space – Bordeaux, France

Developed a deterministic mission-analysis tool solving Lambert’s problem within an iteratively refined grid search, generating a Pareto front (ΔV vs. transfer time) across simultaneous altitude, eccentricity, and inclination corrections; Pareto-dominated the existing sequential-maneuver baseline.

Propagated orbital dynamics via RK4 integration ($\sim 10^{-3}$ m precision) including J2, solar radiation pressure, and atmospheric drag perturbations; implemented a Kalman filter estimating spacecraft inertial position and attitude state.

Validated GNC robustness on a 50-run Monte Carlo rendezvous/inspection campaign: impulsive-thrust mode achieved zero failures (mean ΔV 41.9 m/s, sub-meter final dispersion); root-caused a continuous-thrust attitude PID lag error to 5 software defects, which were corrected.

Ported GNC algorithms from Python to C, cutting runtime $8.2\times$ (1.044 ms $\rightarrow 127.1$ μ s) with matching outputs; integrated onto a hardware-in-the-loop Flatsat testbed (Raspberry Pi 4, then the OBS-VSP-v1.0 flight computer).

Graduate Research Assistant Aug 2026–Present

Aerospace Systems Design Laboratory (ASDL), Georgia Tech

Author user, developer, and advanced-user documentation for a confidential multidisciplinary design and optimization (MDAO) rocket-engine cycle-analysis tool, requiring full comprehension of the underlying codebase.

Engineering Intern – System Design & Architecture Jun–Aug 2025

Agema Space – Bordeaux, France

Developed a Satellite Propulsion Calculator (Python, Java/HTML) computing optimized propulsion configurations from mission parameters.

Selected Projects

Space Campaign Technologies Assessment, ASDL, Georgia Tech – Aug 2026–Present: Georgia Tech-sponsored academic project evaluating architectures and technologies for sustained lunar surface base operations and logistics.

AI for Engineering Data Retrieval (with Gulfstream), Georgia Tech – Metz, Fall 2025: Contributed to an LLM-based knowledge-graph system helping engineers search and interpret technical data.

Competition Propeller Design, ONERA – Lille, Sep 2024–Feb 2025: Designed and manufactured a propeller with a 5-student team (MATLAB, CATIA V5, CFD) for noise and energy efficiency.

Skills

Programming: Python, C, C++ (coursework/personal projects), MATLAB/Simulink, Git, LaTeX

GNC & Mission Design: Orbital propagation & perturbation modeling (J2, drag, SRP), Lambert-problem trajectory optimization, Kalman filtering, Monte Carlo validation, HIL integration

Engineering Software: CATIA V5, SolidWorks, ANSYS

Languages: French (Native), English (TOEIC 930/990)